PROGRAMMING 731 ASSIGNMENT

RICHFIELD
richfield.ac.za

Name and Surname: _____

Student ITS No: _____

Qualification: _____ **Year of Study:** _____ **Semester:** _____

Assignment due date: _____ **Date submitted:** _____

QUESTION	EXAMINER MARKS	MODERATOR MARKS	REMARKS

ASSIGNMENT INSTRUCTIONS

Please tick each box to confirm completion.

Use Times New Roman font, size 12, with 1.5 line spacing throughout the document. Apply Harvard Referencing Style for all citations and references.

For essay-style assignments, please include the following sections:

Table of Contents
Introduction
Main Body (with relevant subheadings)
Conclusion
References

Submit the assignment in PDF format on Moodle. Use the specified cover page provided.
Include a signed declaration of originality.

DECLARATION OF ORIGINALITY:
I hereby declare that this assignment is my own work and has not been copied from any other source except where due acknowledgment is made. I affirm that all sources used have been properly cited and that this submission complies with the institution's policies on academic integrity and plagiarism.

Student Signature: _____ **Date:** _____

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QUESTION 1 (100 MARKS)

1. Assignment Overview

Richfield wants to develop a **Campus Event Management System** that will allow **students and staff** to create, manage, view, and attend events happening on campus.

You are required to design and implement a **Java console-based application** that meets the functional and technical requirements outlined below. The system should simulate **real-world behaviour**, including role-based access control, event capacity management, waitlists, and automated system responses.

2. Functional Requirements (100 Marks)

2.1 User Roles and Access Control (20 Marks)

The system must support **two types of users**:

Staff Users

Staff users must be able to:

- Create events
- Update event details
- Cancel events
- View all registered participants and waitlists

Student Users

Student users must be able to:

- View available events
- Register for events
- Cancel event registration
- View their registration status (**Registered** or **Waitlisted**)

The system must:

- Prompt the user to select a role (**Student or Staff**) at program startup
- Restrict access so that students cannot create, update, or cancel events

2.2 Event Management and State Control (30 Marks)

Staff users must be able to:

- Create new events with the following information:
    - Event ID (unique integer)
    - Event Name
    - Event Date (dd/mm/yyyy)
    - Event Time (HH:mm)
    - Location
    - Maximum Participants
- Update event details (name, time, or location)
- Cancel events
- View all events displaying:
    - Current number of registered participants
    - Number of users on the waitlist
- Sort events by **Event Name** or **Event Date**

Each event must maintain two distinct states:

- Registered participant list
- Waitlist

2.3 Student Registration, Waitlists, and Automation (30 Marks)

The system must allow students to:

- Register for an event using the Event ID
- Be automatically:
    - Registered if space is available
    - Placed automatically on a waitlist if the event is full
- Cancel their registration or waitlist entry
- When a registered student cancels, the first student on the waitlist must be automatically promoted to the event.
- This promotion must be handled in a separate thread to simulate background processing.
- The system must display a notification message confirming the promotion:

  Example: "Registration cancelled. Student S102 has been promoted from the waitlist to the

event."

2.4 Search, Validation, Error Handling, and Persistence (20 Marks)

Search

The system must allow users to search for events by:

- Event Name (partial or full match)
- Event Date

Full event details must be displayed, including registration and waitlist counts.

Validation and Exception Handling

The system must validate:

- Numeric and unique Event IDs
- Correct date and time formats
- Positive maximum participant values
- Non-empty text fields

The system must handle errors gracefully, including:

- Invalid input types
- Non-existent events
- Duplicate registrations

Data Persistence

- Save events, registrations, and waitlists to files
- Load all saved data automatically when the program starts

3. Technical Requirements

The following technical expectations apply **across all sections** of the assignment and will be considered when awarding marks within each functional requirement:

This includes:

- Use of **classes and objects** to model real-world entities (e.g., users and events)
- Use of **inheritance and polymorphism** where appropriate (e.g., different user roles)
- Use of **arrays or Java collection frameworks** to manage multiple records
- Appropriate use of **ArrayLists, Queues, or Maps** for event registrations and waitlists
- **Modular programming** using methods for readability and maintainability

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- Proper **exception handling** and **file input/output**

The choice of specific data structures and overall design approach is left to the student, provided the solution is logical, well-structured, and aligned with OOP principles.

The final application must be fully operational. No marks will be awarded for any application that does not function correctly. Link to demo of the expected system: PRO731_Assignment_System_Functonality.mp4

4. Submission Requirements

Students must submit the following on Moodle:

- A complete **ZIP folder** not bigger than 50mb containing:
    - All Java source files (.java)
    - Any data files used for persistence(eg .txt/.dat)
- The program must:
    - Compile and run successfully
    - Be fully console-based (no GUI)
- Code must be:
    - Clearly structured
    - Properly commented
    - Easy to navigate
    - Copy-and-paste code is strictly prohibited

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